

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 2 of 10

Statements of the Second Law and their Equivalence

Kelvin–Planck statement · PMM2 · Clausius statement · proof of equivalence · corollaries of the second law · reversibility revisited

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2. Equivalence: violating Kelvin–Planck $\Rightarrow$ violating Clausius
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A law of experience

The second law has no single formula. It is a statement of experience that has never been contradicted.

- Two classical statements are in use: one about heat engines (Kelvin–Planck), one about refrigerators (Clausius)

1

The two statements

Kelvin–Planck statement

It is impossible to construct a device that, operating in a cycle, produces no effect other than the raising of a weight and the exchange of heat with a single reservoir.

- No heat engine can convert **all** the heat it receives into work – part must be rejected to a sink at a lower temperature
- A heat engine must exchange heat with **at least two** reservoirs at different temperatures

A process is not a cycle

- An ideal gas expanding isothermally converts all the heat received into work – allowed
- But the gas is left in a different state: this is a **process**, not a cycle
- It is the **continuous, cyclic** conversion of heat entirely into work that is impossible

Perpetual motion machine of the second kind (PMM2)

- A hypothetical engine receiving heat from a **single reservoir** and converting it entirely into work
- Does not violate the first law – it creates no energy
- Would let a ship run on the heat of the ocean, or a power plant on the heat of the atmosphere – no fuel, no sink
- Kelvin–Planck in short: **a PMM2 is impossible**

KEY POINT PMM1 (Unit I) would create energy and violates the first law; PMM2 creates nothing but violates the second law.

Clausius statement

It is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a cooler body to a hotter body.

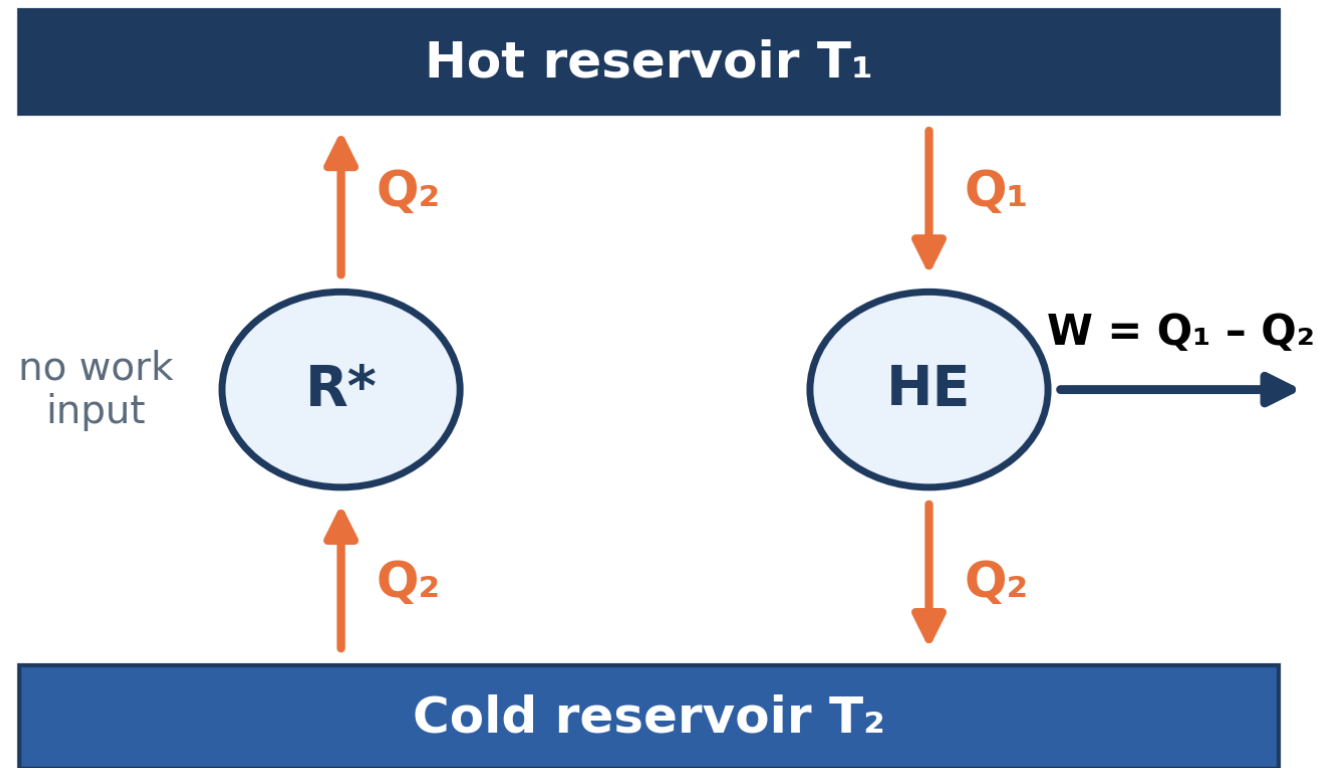
- Heat flows of its own accord only from hot to cold; to make it flow "uphill" a refrigerator – and a work input – is needed
- Every refrigerator does move heat from cold to hot – but never **without some other effect** (the consumption of work)

2

Equivalence of the two statements

A violation of either implies a violation of the other

Violating Clausius $\Rightarrow$ violating Kelvin–Planck

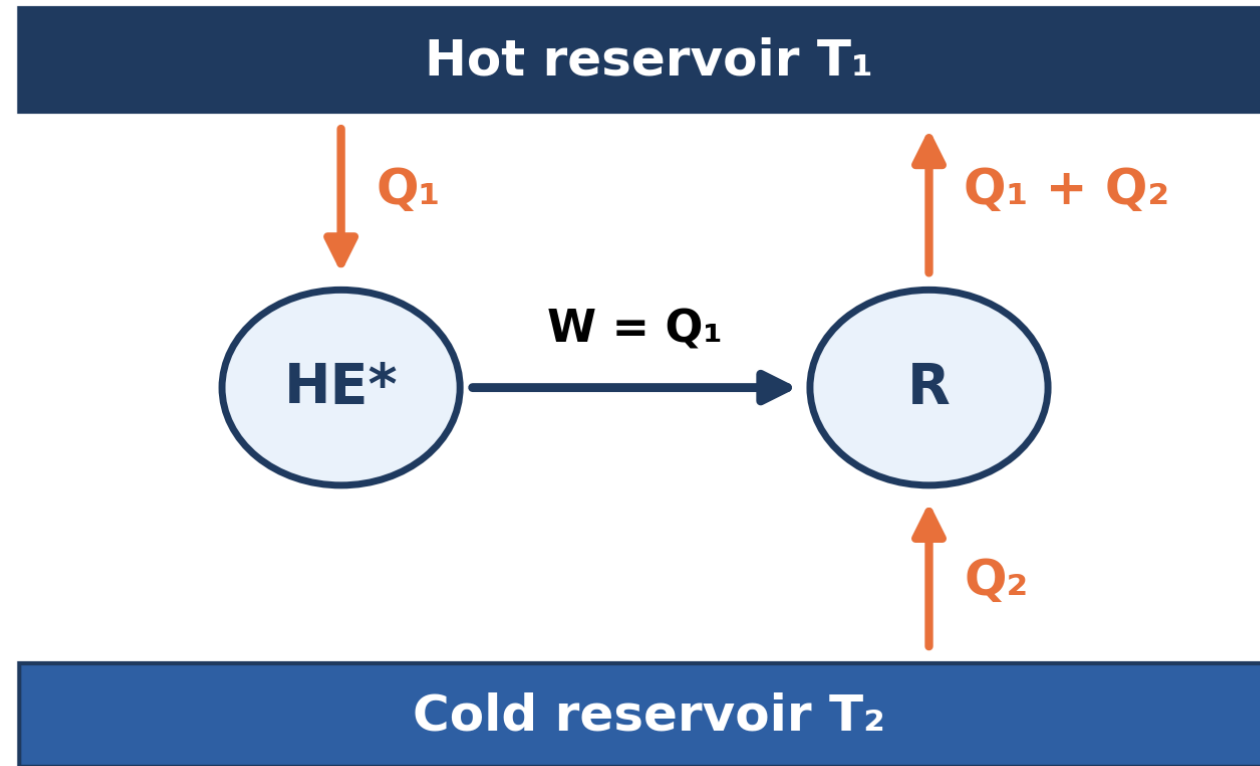


R^* moves Q_2 from cold to hot with no work; combined with an ordinary engine HE the pair is a PMM2

Violating Clausius $\Rightarrow$ violating Kelvin–Planck

1. Suppose device R^* transfers Q_2 from the cold reservoir at T_2 to the hot reservoir at T_1 **with no work** (Clausius violator)
2. Let an ordinary engine HE take Q_1 from the hot reservoir, deliver $W = Q_1 - Q_2$ and reject Q_2 to the cold reservoir
3. The cold reservoir receives Q_2 from HE and gives Q_2 to R^* – it is unaffected and can be removed
4. The combined plant takes $Q_1 - Q_2$ from the hot reservoir **alone** and delivers equal work – a PMM2

Violating Kelvin–Planck $\Rightarrow$ violating Clausius



HE^* converts Q_1 entirely into work, which drives refrigerator R ; the net effect moves Q_2 from cold to hot with no external work

Violating Kelvin–Planck $\Rightarrow$ violating Clausius

1. Engine HE* takes Q_1 from the hot reservoir and converts it **completely** into work $W = Q_1$ (Kelvin–Planck violator)
2. This work drives an ordinary refrigerator R: absorbs Q_2 from the cold reservoir, rejects $Q_1 + Q_2$ to the hot
3. Hot reservoir: net gain Q_2 ; cold reservoir: loses Q_2 ; the work is internal
4. Net effect: Q_2 moved from cold to hot with **no external work** – Clausius violated

KEY POINT Each violation implies the other, so the two statements are equivalent expressions of one law.

Corollaries of the second law

1. No engine between two given reservoirs can be more efficient than a reversible engine between the same reservoirs (Carnot theorem)
2. All reversible engines between the same two reservoirs have the same efficiency
3. That efficiency depends only on the two reservoir temperatures – basis of the absolute temperature scale
4. The COP of a reversible refrigerator or heat pump is the maximum possible and depends only on the two temperatures
5. The entropy of an isolated system never decreases

3

Reversibility revisited

What makes a process irreversible

Reversible and irreversible processes

- **Reversible:** can be reversed so that both system and surroundings return to their initial states – no trace anywhere
- Must be quasi-static (infinitely slow, through equilibrium states) **and** free of dissipative effects
- An idealisation – **every real process is irreversible**

Two groups of causes

Mechanical irreversibility – dissipation

- Friction in solids and fluids
- Electric current through a resistance
- Inelastic deformation, magnetic hysteresis
- Work becomes internal energy and cannot be recovered as work

Lack of equilibrium – finite driving forces

- Heat transfer through a finite ΔT
- Free expansion into a vacuum
- Mixing of gases or of streams at different temperatures
- Spontaneous chemical reaction; throttling

Internal and external irreversibility

- **Internal** – within the system: friction, mixing, unrestrained expansion
- **External** – at the boundary: heat transfer across a finite temperature difference with the surroundings
- **Internally reversible** process: free of internal irreversibility; a reversible process is free of both
- Reversible processes deliver the **maximum work** (or need the minimum) – the standard for real processes; the Carnot cycle is built from them

Summary

- Kelvin–Planck: no cyclic device produces work from a single reservoir – no PMM2
- Clausius: no cyclic device moves heat from cold to hot with no other effect
- Violating either statement builds a violator of the other – equivalent
- Corollaries: Carnot theorem, absolute temperature scale, maximum COP, entropy of an isolated system never decreases
- Irreversibility: dissipation, or finite driving forces (ΔT , free expansion, mixing, throttling)
- Reversible processes give the maximum work

Next: The Carnot cycle

- The four reversible processes on p–V and T–s diagrams
- Carnot efficiency $\eta = 1 - T_2/T_1$; the reversed Carnot cycle and its COPs
- Why the Carnot cycle cannot be built – and why it still matters